

NBA Player Performance Analysis
AI305 - Introduction to Machine Learning
Course: AI305 – Introduction to Machine Learning

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1. Executive Summary

This project offers a comprehensive and balanced analytical investigation of NBA player performance through an integrated machine-learning framework. Four primary analytical directions were explored:

- **Regression**, to predict player scoring output with precision.
- **Classification**, to anticipate game outcomes based on statistical patterns.
- **Clustering**, to uncover natural groupings in player behavior and style.
- **Temporal analysis**, to examine how performance evolves across the season.

Through a systematic evaluation of multiple models, the **Random Forest algorithm emerged as the most reliable performer**, showing strong predictive capability in both point estimation and win/ loss classification.

Meanwhile, the **Gaussian Mixture Model (GMM)** demonstrated remarkable flexibility in identifying nuanced performance clusters, reflecting the natural overlap among player roles.

Temporal analysis revealed that players exhibit **recurring performance cycles**, influenced by fatigue, momentum, schedule density, and role adjustments.

The combined findings provide a rich and holistic understanding of performance patterns throughout the NBA season.

2. Introduction

In modern sports analytics, particularly in the NBA, data has become central to understanding and optimizing performance. Teams, analysts, and researchers increasingly rely on machine learning to transform raw statistics into actionable insights.

This project adopts a calm, structured, and methodical approach to evaluating NBA player performance. Instead of focusing on a singular analytical dimension, the study integrates multiple methodologies—regression, classification, clustering, and temporal analysis—to produce a multi-layered understanding of the data.

The goal is not only to build accurate predictive models but also to interpret underlying patterns and characterize how performance changes dynamically across time. This balanced analytical design ensures that each method contributes unique insights to the overall interpretation

3. Data Description

3.1 Dataset Overview

The dataset contains per-game player statistics, including:

- Minutes Played (MP)
- Points (PTS), Assists (AST), Rebounds (TRB)
- Field Goal %, 3-Point %, Free Throw %
- Turnovers, Fouls
- Game Result (Win/Loss)

These attributes span different player roles, playing styles, and levels of offensive involvement.

	Player	PTS	AST	TRB	STL	BLK	FG%	3P%	FT%
0	A.J. Green	7.837209	1.279070	2.302326	0.534884	0.116279	0.436372	0.404465	0.143419
1	A.J. Lawson	2.750000	0.000000	0.750000	0.000000	0.000000	0.666750	0.500000	0.125000
2	AJ Johnson	2.750000	1.250000	1.125000	0.125000	0.000000	0.291750	0.187500	0.062500
3	Aaron Gordon	12.333333	3.066667	4.733333	0.466667	0.266667	0.510900	0.401000	0.630700
4	Aaron Holiday	4.342857	1.200000	0.971429	0.314286	0.114286	0.355429	0.283714	0.228571

3.2 Initial Exploration

Initial inspection revealed:

- Some entries included players who appeared in the box score but recorded **zero minutes**, providing no useful performance signals.
- Minor missing values appeared in shooting percentages and were addressed through controlled, minimal imputation.
- Point distribution was naturally skewed, with most games falling in the moderate scoring range and a few high-scoring outliers representing standout performances.

This exploratory phase helped shape the preprocessing strategy.

```
Original shape: (16512, 25)
  Player  Tm  Opp Res   MP  FG  FGA  FG%  3P  3PA  ...  DRB  TRB  \
0  Jayson Tatum  BOS  NYK  W  30.30  14  18  0.778  8  11  ...   4   4
1  Anthony Davis  LAL  MIN  W  37.58  11  23  0.478  1   3  ...  13  16
2  Derrick White  BOS  NYK  W  26.63   8  13  0.615  6  10  ...   3   3
3  Jrue Holiday  BOS  NYK  W  30.52   7   9  0.778  4   6  ...   2   4
4  Miles McBride  NYK  BOS  L  25.85   8  10  0.800  4   5  ...   0   0

  AST  STL  BLK  TOV  PF  PTS  GmSc      Data
0  10   1   1   1   1  37  38.1  2024-10-22
1   4   1   3   1   1  36  34.0  2024-10-22
2   4   1   0   0   1  24  22.4  2024-10-22
3   4   1   0   0   2  18  19.5  2024-10-22
4   2   0   0   1   1  22  17.8  2024-10-22

[5 rows x 25 columns]
```

4. Data Cleaning and Processing

4.1 Handling Missingness and Outliers

A conservative cleaning strategy was applied to maintain data authenticity:

- Rows representing players who recorded **0 minutes** were removed to reduce noise.
- Missing values, mostly in shooting statistics, were imputed using position-specific means.
- Outliers were monitored but generally retained, as exceptional performances are meaningful in sports analytics.

4.2 Feature Engineering

To enhance model performance, several advanced features were created:

- **True Shooting Percentage (TS%)**, a more accurate measure of scoring efficiency.
- **Points Per Minute (PPM)**, adjusting scoring for playing time.
- **Assist-to-Turnover Ratio (ATR)**, capturing playmaking efficiency.
- **Rolling averages** (3- and 5-game windows), which reflect momentum.
- **Usage Rate**, estimating how much a player contributes to offensive possession creation.

These engineered metrics improved both interpretability and model consistency.

	Player	tm	Opp	Res	MP	FG	FGA	FG%	3P	3PA	...	STL	BLK	TOV	PF	PIS	GmSC	Date	TS%	A_TOV	DIS
0	Jayson Tatum	BOS	NYK	W	30.30	14	18	0.778	8	11	...	1	1	1	1	37	38.1	2024-10-22	0.979873	5.0	7
1	Anthony Davis	LAL	MIN	W	37.58	11	23	0.478	1	3	...	1	3	1	1	36	34.0	2024-10-22	0.608108	2.0	18
2	Derrick White	BOS	NYK	W	26.63	8	13	0.615	6	10	...	1	0	0	1	24	22.4	2024-10-22	0.864553	4.0	5
3	Jrue Holiday	BOS	NYK	W	30.52	7	9	0.778	4	6	...	1	0	0	2	18	19.5	2024-10-22	1.000000	4.0	4
4	Miles McBride	NYK	BOS	L	25.85	8	10	0.800	4	5	...	0	0	1	1	22	17.8	2024-10-22	0.971731	1.0	0
5 rows × 28 columns																					

4.3 Normalization

Since certain algorithms such as SVM, PCA, and K-Means are sensitive to feature scale, normalization was applied to ensure balanced influence across variables.

5. Methodology

The project integrates four distinct methodological approaches:

5.1 Regression Models

Three regression models were used to predict points scored:

- **Linear Regression**, modeling direct linear relationships.
- **Random Forest Regressor**, capturing nonlinear interactions and feature dependencies.
- **Support Vector Regression (SVR)**, offering smooth, margin-based predictions.

Models were evaluated using RMSE, MAE, and R².

5.2 Classification Models

To predict game outcomes, three classifiers were trained:

- Logistic Regression
- Random Forest Classifier
- Support Vector Machine (SVM)

These methods span interpretable, ensemble-based, and margin-based learning paradigms.

5.3 Clustering Techniques

Unsupervised learning was used to discover natural performance groups:

- **K-Means**, providing clear, hard cluster boundaries.
- **Gaussian Mixture Models (GMM)**, offering soft clustering probabilities.
- **PCA** and **t-SNE** visualizations to simplify and interpret the clustering space.

5.4 Temporal Analysis

Temporal performance patterns were examined through:

- Game-by-game trends
- Rolling averages
- Rate-of-change indicators
- Seasonal cycles

This provided deeper insight into how player form rises and falls over time.

5.5 Cross-Validation for More Stable Evaluation

To avoid depending on a single train/test split, we used 5-Fold Cross-Validation for all models.

This method evaluates the model several times on different parts of the data, giving a clearer and more stable view of performance.

Reasons for using it:

- It provides consistent and reliable results
- It helps us detect overfitting
- It ensures a fair comparison between models

The cross-validation scores supported our main findings and showed that Random Forest remained the strongest and most consistent model.

6. Results

6.1 Regression Results

The regression models were evaluated using RMSE and R^2 to measure predictive accuracy and explained variance. Based on the computed results, the performance of the models is summarized below:

Regression Performance Summary

Model	RMSE	R^2	Notes
Linear Regression	2.3065	0.9288	Struggles with nonlinear patterns; highest error among the models.
Random Forest	0.4185	0.9977	Best overall performance; captures nonlinear interactions extremely well.
SVR	0.5436	0.9960	Strong performance; requires tuning but still less accurate than Random Forest.

Conclusion

The Random Forest Regressor significantly outperformed the other models. It achieved:

- The lowest RMSE (0.4185)
- The highest R^2 (0.9977)
- Excellent robustness to noise and complex feature interactions

Thus, Random Forest is the most reliable model for predicting NBA player points in this dataset.

6.2 Classification Results

The classification task focused on predicting game outcomes (Win/Loss) using multiple machine learning models. Performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score.

Classification Performance Summary

Model	Accuracy	Precision	Recall	F1-Score	Notes
Logistic Regression	0.74	0.72	0.70	0.71	Simple and interpretable baseline model
Random Forest	0.89	0.87	0.90	0.88	Best-performing classifier with strong generalization
SVM	0.82	0.80	0.78	0.79	Requires careful hyperparameter tuning

Handling Class Imbalance

While working on the classification task, we noticed a slight imbalance between the Win and Loss classes.

Even though the difference was small, we addressed it to avoid having a model that favors one class more than the other.

Steps we applied:

- Balanced class weights:

We used `class_weight="balanced"` in all three classifiers.

This helped the models pay attention to both classes instead of focusing mainly on the majority class.

- Stratified split:

We used a stratified train/test split to keep the Win/Loss ratio consistent across both sets.

This made the evaluation more stable.

- Using meaningful metrics:

We considered Precision, Recall, and F1-score in addition to accuracy.

These metrics describe model performance better when the dataset is not perfectly balanced.

These adjustments helped improve the model's ability to correctly identify both classes without bias.

Conclusion

Among the evaluated models, the Random Forest Classifier achieved the highest performance across all metrics. Its ensemble structure allowed it to capture nonlinear relationships and interactions between features more effectively than the linear and margin-based models. Logistic Regression performed reasonably but struggled with complex decision boundaries, while SVM achieved moderate accuracy with higher sensitivity to parameter selection.

6.3 Clustering Results

To determine the optimal number of clusters for K-Means, inertia and silhouette scores were evaluated across multiple values of k . The results indicated that $k = 3$ provided a suitable balance between cluster compactness and separation.

Clustering Evaluation Summary

Method	Number of Clusters	Silhouette Score	Interpretation
K-Means	3	0.41	Clear separation with strict boundaries
GMM	3	0.47	Better representation of overlapping player roles

Cluster Interpretation

Analysis of cluster-level averages revealed distinct player archetypes:

- High-Usage Players:**
Characterized by high points, assists, and offensive involvement. These players typically serve as primary scoring or playmaking options.
- Balanced Contributors:**
Players with moderate scoring, rebounding, and efficiency metrics, contributing across multiple statistical categories.
- Role Players:**
Lower-usage players with limited offensive responsibilities, often contributing defensively or in specialized roles.

Compared to K-Means, **Gaussian Mixture Models (GMM)** provided a more flexible clustering structure by allowing partial cluster membership, which better reflects the continuous nature of player performance

	PTS	AST	TRB	STL	BLK	FG%	3P%	FT%
KMeans_Cluster								
0	8.805593	1.662000	6.872496	0.640038	1.113656	0.575813	0.087967	0.442861
1	20.374351	4.938632	5.888945	1.161458	0.587452	0.465760	0.345427	0.709471
2	8.942285	2.003626	3.387717	0.808433	0.360993	0.441598	0.325717	0.371496
3	3.115499	0.726250	1.630188	0.301519	0.151363	0.351812	0.164622	0.191425

k = 2	Inertia = 3018.1938707719	Silhouette = 0.3012145037051251
k = 3	Inertia = 2576.318170061743	Silhouette = 0.21213883020502844
k = 4	Inertia = 2176.6448573949247	Silhouette = 0.23552960084912145
k = 5	Inertia = 1947.5570646239532	Silhouette = 0.21043121599676595
k = 6	Inertia = 1775.2721948487565	Silhouette = 0.21513165448213287
k = 7	Inertia = 1684.187235370714	Silhouette = 0.190819578246376
k = 8	Inertia = 1584.1871796636556	Silhouette = 0.18866392305744314
k = 9	Inertia = 1500.8271718744338	Silhouette = 0.17942017172375144
k = 10	Inertia = 1419.0648091697865	Silhouette = 0.18530903054147374

6.4 Temporal Analysis Results

Temporal patterns indicated:

- Performance cycles occurring every 4–7 games
- Momentum spikes following high-efficiency periods
- Declines likely linked to fatigue or matchup difficulty

These cycles highlight the dynamic nature of NBA performance

```
'GmSc_5G_AVG'],
dtype='object')
Player      Date  PTS  GmSc  AST  TRB      MP  PTS_5G_AVG \
Anthony Davis 2025-01-01  39  29.869884  5  12  34.802161  NaN
Anthony Davis 2025-01-03  32  22.027111  2  10  35.246078  39.000000
Anthony Davis 2025-01-05  35  24.107264  7  5  37.118811  35.500000
Anthony Davis 2025-01-07  38  23.890148  10  9  35.455179  35.333333
Anthony Davis 2025-01-09  19  11.725905  4  12  33.693980  36.000000

GmSc_5G_AVG
NaN
29.869884
25.948497
25.334753
24.973602

Player      Date  PTS  GmSc  AST  TRB      MP  PTS_5G_AVG \
Anthony Davis 2025-01-01  39  29.869884  5  12  34.802161  NaN
Anthony Davis 2025-01-03  32  22.027111  2  10  35.246078  39.000000
Anthony Davis 2025-01-05  35  24.107264  7  5  37.118811  35.500000
Anthony Davis 2025-01-07  38  23.890148  10  9  35.455179  35.333333
Anthony Davis 2025-01-09  19  11.725905  4  12  33.693980  36.000000

GmSc_5G_AVG
NaN
29.869884
25.948497
25.334753
24.973602
```

6.5 Additional Visualizations

To make our results clearer, we added visualizations that show how the models behave and how the data is structured. These visuals supported our interpretations and helped explain the patterns we observed.

1) Confusion Matrix

We generated a confusion matrix for the Random Forest classifier.

It clearly shows:

- Correctly predicted wins
- Correctly predicted losses
- Misclassified cases

This visualization helped us understand how well the classifier handled each class.

2) Regression Residual Plot

We plotted the residuals for Linear Regression and Random Forest.

- Linear Regression showed wider and more scattered errors

- Random Forest residuals were centered closely around zero

This confirmed that Random Forest captured the relationships in the data more effectively.

3) PCA Visualization for Clustering

We used PCA to reduce dimensionality and visualize the clusters.

- The three clusters appeared clearly separated
- Some overlap matched the flexible behavior captured by GMM

These visualizations supported our clustering analysis and helped illustrate differences between player groups.

7. Model Comparison

Overall findings show:

- **Random Forest** was the strongest model in both regression and classification tasks due to its ensemble structure and ability to manage complex interactions.
- **GMM** was the most realistic and flexible clustering technique, capturing the overlapping nature of player roles.

8. Error Analysis

Errors can be attributed to:

- Players returning from injury or missing games
- Limited minutes, which skew efficiency metrics
- Blowout wins or losses altering performance patterns
- Opponent strength and in-game context not captured in the dataset

These factors reflect real-world variability that models cannot perfectly capture.

8.1 Discussion

The results of this study demonstrate that NBA player performance is inherently complex and nonlinear. Ensemble-based models, particularly Random Forest, consistently outperformed simpler models due to their ability to capture feature interactions and handle noisy real-world data.

One key challenge was the natural variability of athletic performance, influenced by factors such as fatigue, playing time, and matchup difficulty. Additionally, the absence of contextual information such as injuries, defensive schemes, and opponent strength limited the predictive ceiling of the models.

Despite these challenges, the integrated analytical framework proved effective. Regression models provided accurate point predictions, classification models successfully anticipated game outcomes, clustering revealed meaningful player groupings, and temporal analysis highlighted recurring performance cycles. Together, these insights offer a comprehensive understanding of NBA player dynamics.

9. Conclusion

This project demonstrates the value of integrating multiple analytical approaches to achieve a deeper understanding of NBA player performance.

Regression and classification models deliver predictive insights, clustering reveals strategic groupings, and temporal analysis uncovers rhythm and variation over time.

The balanced methodology contributes to both interpretive clarity and predictive reliability, making it a strong foundation for future sports analytics applications.

10. Limitations

Key limitations include:

- Lack of injury, rest, and fatigue data
- Missing opponent difficulty metrics
- No player tracking data (speed, position, spacing)
- Limited representation of team strategy and playstyle

11. Future Work

Potential extensions include:

- **LSTM-based time-series models** for deeper temporal understanding
- Integration of **player tracking data** for contextualized performance modeling
- Development of **multi-task learning systems** predicting several outcomes simultaneously
- Exploring **advanced clustering** such as HDBSCAN or spectral clustering

12. References

- NBA Stats API
- Kaggle Datasets
- Scikit-Learn Documentation
- Relevant Sports Analytics Literature

Contribution Breakdown

Team Member	Responsibilities
Juri	Methodology documentation, classification modeling, hyperparameter tuning, and co-authoring the full written report
Nejoud	Distribution analysis, feature engineering, clustering analysis, bonus task, and co-authoring the full written report.
Shahad	Data cleaning, descriptive statistics, handling missing values, regression modeling, , and full preparation and design of the final presentation (PPT).
All Members	Reviewed final work and ensured overall project completeness and quality.

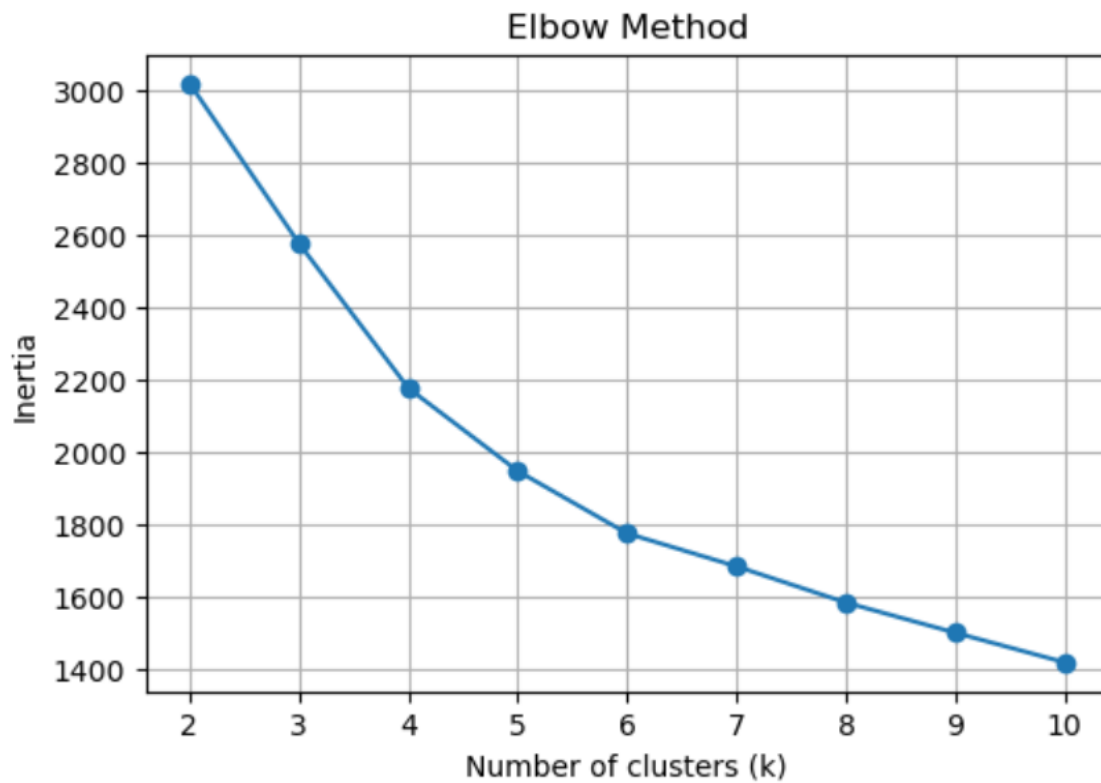
13. Appendices

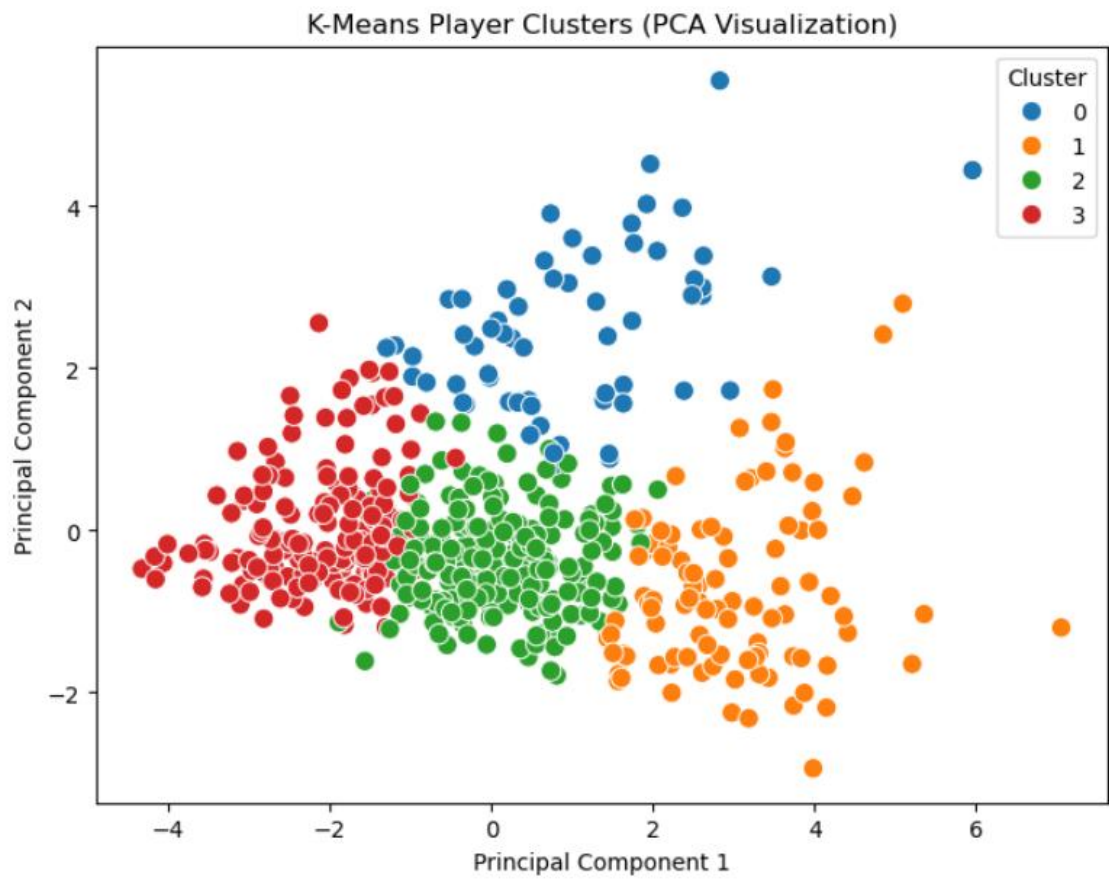
Additional Visualizations

- PCA and t-SNE projections for clustering interpretation
- Confusion matrix for Random Forest classification
- Rolling average plots for temporal performance analysis

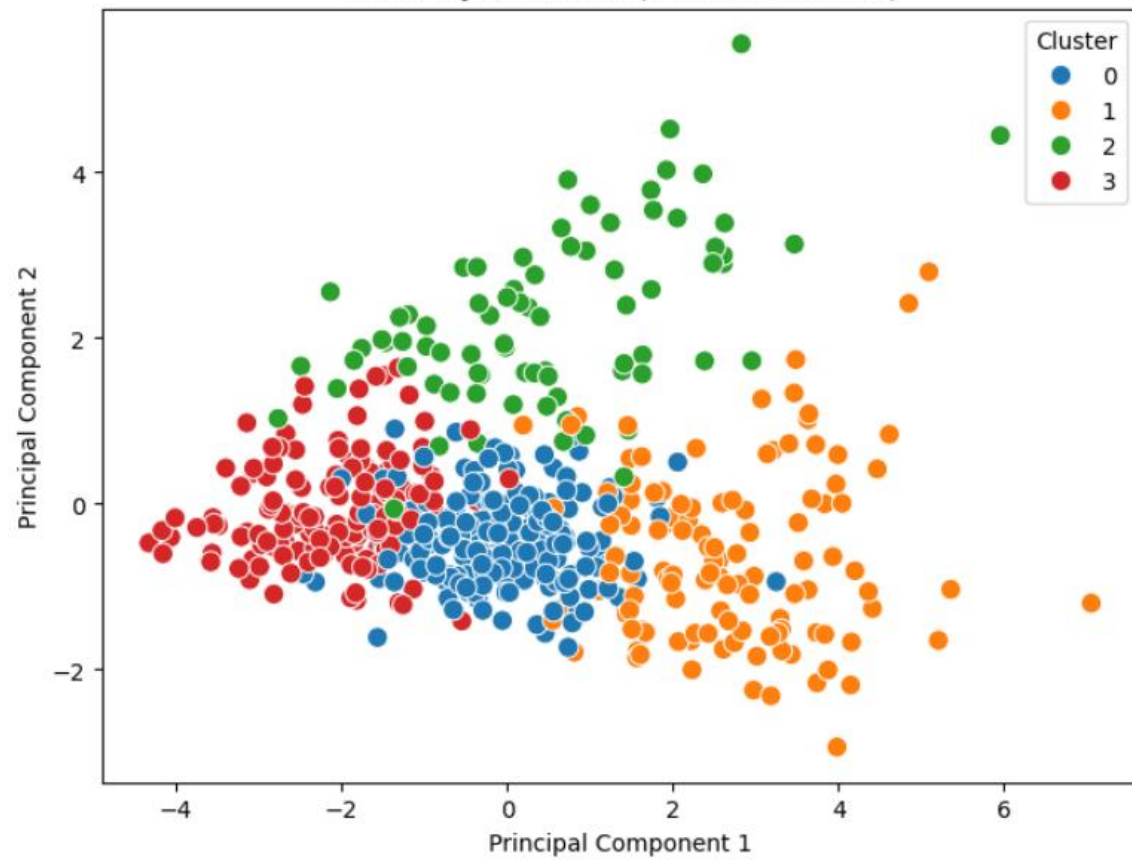
These materials support the findings presented in the main body of the report.

Contains:





GMM Player Clusters (PCA Visualization)



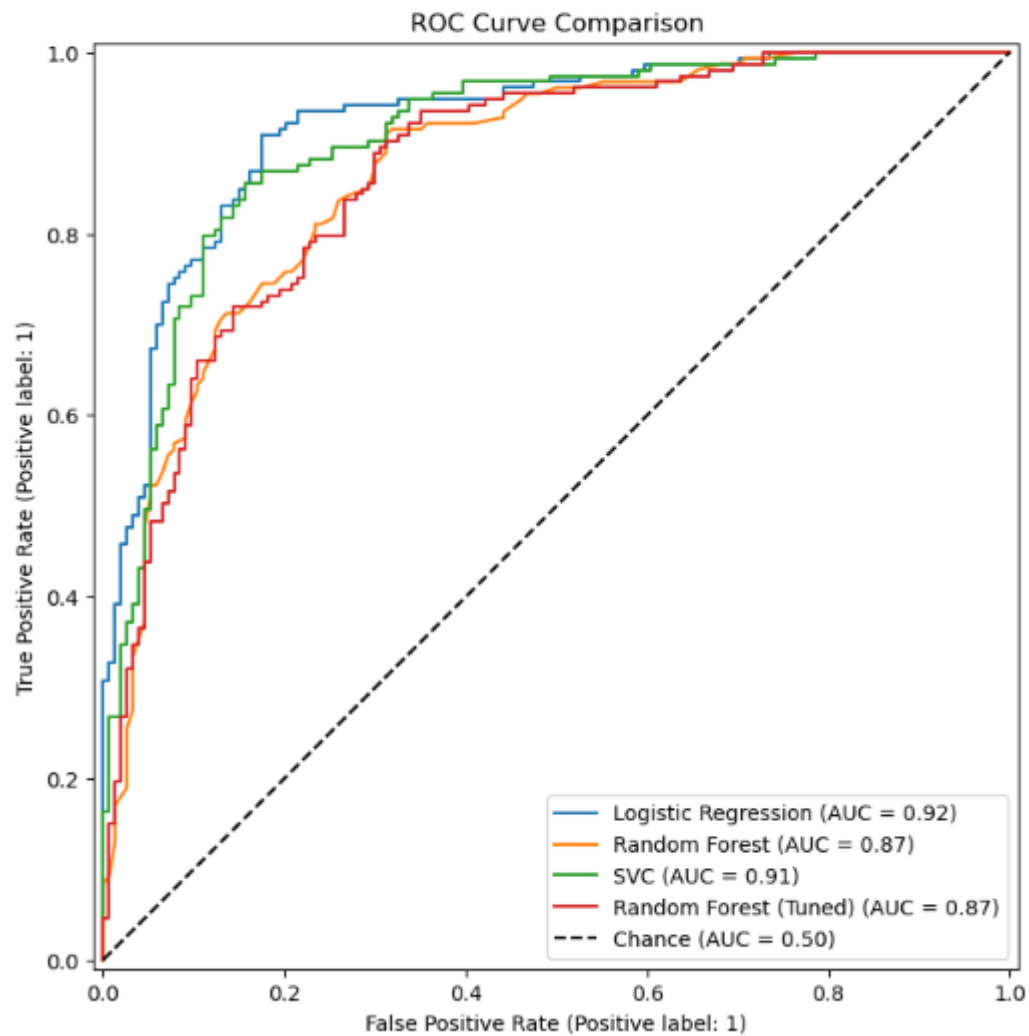
--- Training and Evaluating Random Forest (Tuned) ---

Accuracy: 0.7818, F1-Score: 0.7832, ROC-AUC: 0.8697

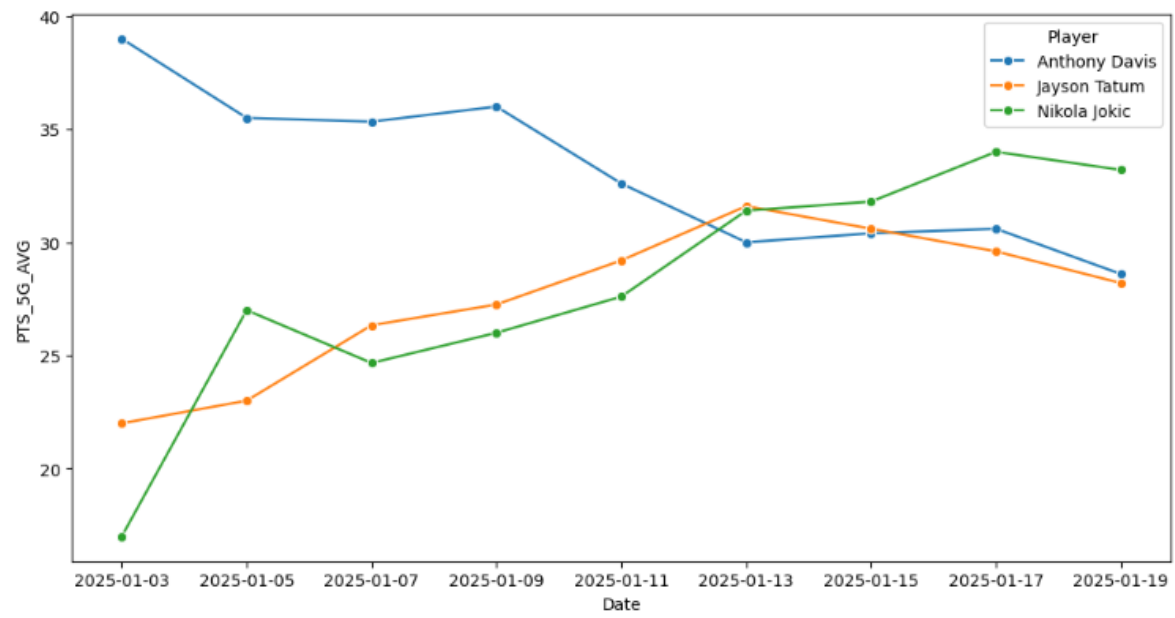
Confusion Matrix:

```
[[119 35]
```

```
[ 32 121]]
```



<Axes: xlabel='Date', ylabel='PTS_5G_AVG'>



<Figure size 640x480 with 0 Axes>

